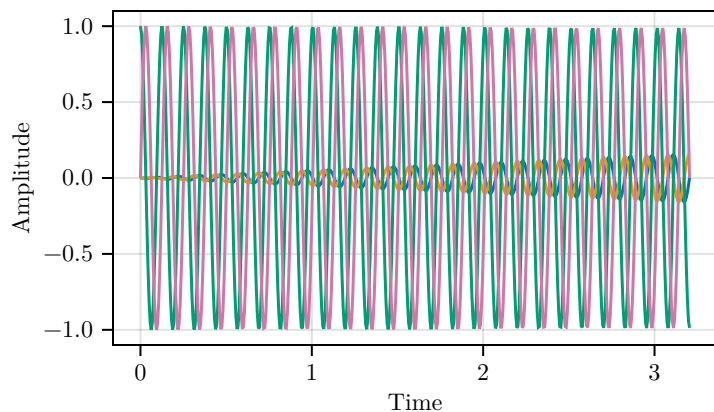
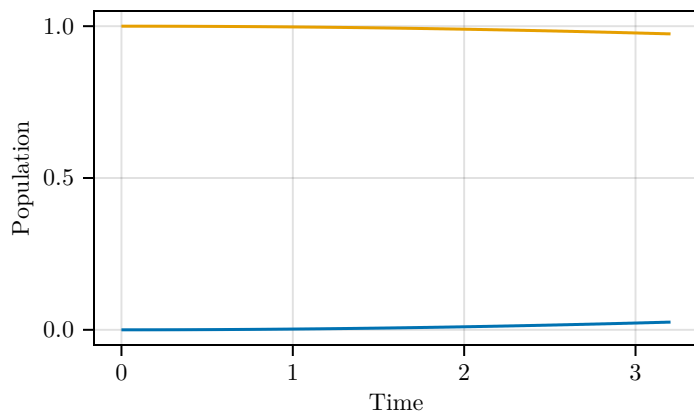


Rabi Oscillation Solutions: $\dot{u} = -i\left(\frac{\omega_0}{2}\sigma_z + \Omega\cos(\omega t)\sigma_x\right)u$, $\omega_0 = 100.0$, $\Omega = 0.1$, $\omega = 100.0$

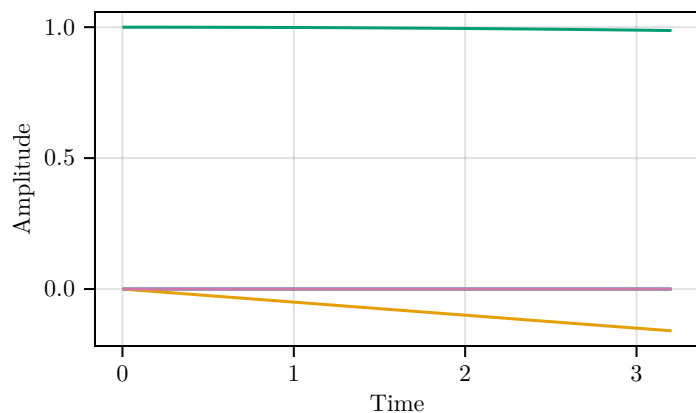
Lab frame : complex amplitudes



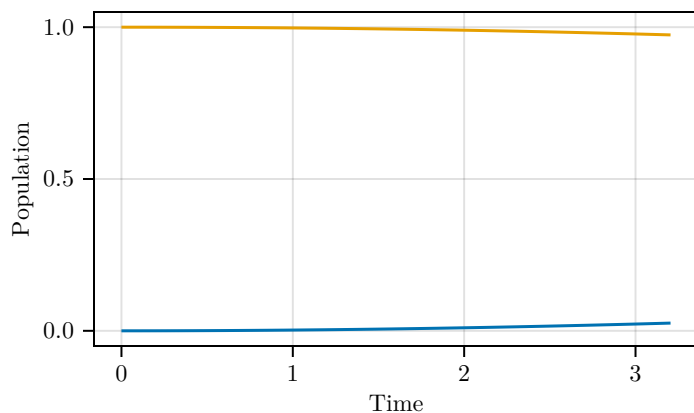
Lab frame : populations



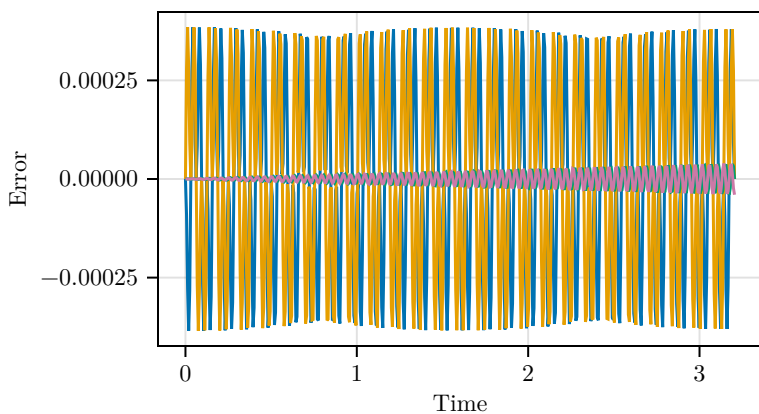
Rotating frame (RWA) : complex amplitudes



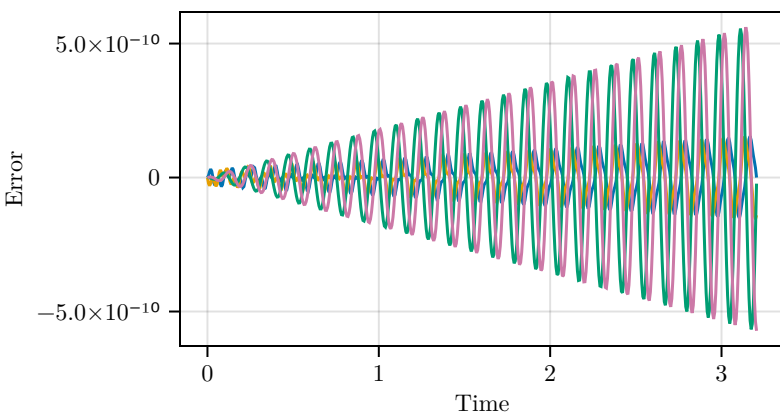
Rotating frame (RWA) : populations



Error : rotating frame (in lab) vs lab frame



Lab frame : Filon error ($\omega \neq 0$) - ($\omega = 0$)



Component

— $\text{Re}(u_1)$ — $\text{Im}(u_1)$ — $\text{Re}(u_2)$ — $\text{Im}(u_2)$